
Breaking the Curse of Repulsion: Distributionally Robust Policy Optimization for Off-Policy Learning

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Abstract

Negative feedback is a central resource in policy optimization: it suppresses known bad behavior and can move a policy beyond the solution obtained from positive examples alone. Yet the same feedback can become destructive when historical negative actions remain in offline data, replay, or stale trajectories after the learner has moved away from them. We identify the missing mechanism by separating sample badness from learner-relative distance or rarity, showing that policy remoteness independently amplifies negative influence. We then develop Repulsive Dynamics, which characterizes a transition from the Positive-only target, through stable extrapolation, to a feasibility boundary and the loss of a finite equilibrium. Based on this analysis, Distributionally Robust Policy Optimization (DRPO) reweights the aggregate negative term with a policy-relative exponential envelope, retaining useful local repulsion while suppressing its far-field tail. Controlled continuous and categorical environments isolate the source and causal transmission of the effect; registered Hopper/D4RL and Countdown experiments are reserved for external validation under learned critics and shared-parameter sequence policies, with formal results remaining TBD until terminal audit. The resulting framework tests a general principle: policy optimization should control far-field negative influence rather than remove negative feedback as a whole.

1. Introduction

Policy optimization learns not only from successful actions but also from actions that should become less likely. Positive updates reinforce observed successes, whereas negative updates suppress known bad modes and shape the boundary between acceptable and unacceptable behavior. This

distinction matters because a learner trained only on positive examples is naturally pulled toward the empirical positive target. When negative repulsion is balanced against that attraction, however, the policy can settle beyond the Positive-only target and improve in regions that were not directly demonstrated by the positive data. The central problem is therefore not whether negative feedback should be used, but how to preserve its policy-improvement value without allowing the same repulsion to become dynamically excessive.

The difficulty arises when the data are historical. A negative action may initially lie near the learner and provide relevant local boundary information, but offline logs, replay buffers, stale actors, and asynchronous trajectories can continue to reuse that action after the policy has moved away. Its learner-relative distance or rarity then increases while its negative label continues to generate updates. For Gaussian policies, the score can grow with standardized distance; for categorical policies, the direct-logit score is bounded, yet repeated negative updates continue to reduce the action's log-odds. Historical reuse can therefore transform locally useful feedback into persistent far-field repulsion.

Existing methods already regulate negative updates in several ways: positive-only objectives remove them, global coefficients reduce their scale, clipping limits individual ratios, behavior constraints restrict support movement, low-probability controls target rare events, and quality filtering removes selected data. These mechanisms can improve stability, but they do not explain why the same negative feedback is useful near the learner and destructive after it becomes far field. The question is difficult to answer from ordinary logs because low reward, large negative advantage, rarity, and policy distance are typically correlated. We therefore isolate policy remoteness while matching context, action quality or semantic role, reward, negative-advantage magnitude, sample count, base coefficient, and policy stage. This control makes distance or rarity an independently testable amplifier rather than a proxy for samples that are simply worse.

Repulsive Dynamics explains the full transition at the level of aggregate policy updates. Positive-only optimization targets the positive moment. A moderate negative contribution

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shifts the finite equilibrium beyond that target, yielding stable extrapolation. As the negative contribution becomes stronger or more outward, the signed target approaches the feasible boundary of the policy family; beyond that boundary, a finite equilibrium no longer exists. Gaussian and categorical policies share this aggregate phase structure, although they express it differently: Gaussian policies can exhibit distance-amplified scores and coupled mean–support dynamics, whereas categorical policies exhibit bounded but persistent probability suppression toward the simplex boundary.

DRPO acts directly on the negative term identified by the theory. It replaces the raw negative contribution with a policy-relative exponential weight, so that nearby negative actions retain substantial influence while increasingly remote actions are attenuated. Under finite-order score growth, the weighted far-field gradient converges to zero. This design is naturally interpreted as distributional control of the empirical actor update: quality-based selection and learner-relative remoteness are distinct axes and are evaluated separately rather than being conflated. The method therefore preserves the local negative feedback responsible for extrapolation while preventing the far-field tail from dominating the aggregate update.

We evaluate this account through four questions. First, do far-field or rare negative updates become disproportionately influential in realistic policy learning? Second, when badness is matched, does policy remoteness independently generate large negative influence and causally transmit into drift or boundary events? Third, does training follow the phase sequence predicted by Theorem 4.1, and can DRPO preserve useful repulsion while preventing the destructive regime under fair gradient budgets? Fourth, does the resulting control improve external tasks? C-U1 and D-U1 provide controlled continuous and categorical identification, whereas registered Hopper/D4RL and Countdown experiments are assigned the external-validation role and remain TBD until formal terminal audit. The paper contributes a theory of Repulsive Dynamics, matched quality–distance and quality–rarity causal identification, DRPO control of the theorem-level negative term, and a pre-registered external–controlled–external evidence chain.

2. Related Work

Advantage-weighted and filtered policy-learning methods use sample quality to emphasize successful behavior or reduce harmful updates (Peng et al., 2019; Kostrikov et al., 2021). Other work on failures and negative reinforcement shows that suppressing known bad behavior can improve decision boundaries, diversity, or long-horizon performance. We build on this evidence rather than treating all negative data as noise. Our focus is the dynamical transition: the

same negative action can be informative when it is near the current learner and excessive after repeated off-policy reuse moves it into the far field.

Off-policy algorithms commonly use importance correction, clipping, trust regions, or replay design to control distribution mismatch (Schulman et al., 2017; Haarnoja et al., 2018). Recent analyses of stale or low-probability actions emphasize that rare negative events can dominate updates in shared-parameter policies. Repulsive Dynamics complements these views by following a negative action across repeated reuse: learner-relative remoteness changes its score contribution, the contributions aggregate, and the signed target can move to a policy boundary where a finite equilibrium is lost.

Conservative and implicit offline-RL methods control extrapolation by pessimistic value estimation, in-support dynamic programming, or explicit behavior regularization (Kumar et al., 2020; Kostrikov et al., 2021; Fujimoto et al., 2019). These approaches address central offline-RL failures and provide strong external baselines. DRPO studies a different but compatible object: the signed actor update itself. Its purpose is to retain useful local negative feedback while attenuating the learner-relative far-field component that can dominate the policy dynamics.

3. Problem Setup

Let ν denote the empirical update distribution induced by offline data, replay, or stale trajectories. We study the signed actor field

$$\mathbf{F}(\theta) = \mathbb{E}_{(s,a) \sim \nu} \left[\hat{A}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s) \right]. \quad (1)$$

Writing $A^+ = \max(\hat{A}, 0)$ and $A^- = \max(-\hat{A}, 0)$ gives $\mathbf{F} = \mathbf{F}^+ - \mathbf{F}^-$. For a negative sample,

$$\|g_i^-\| = A_i^- \|\nabla_{\theta} \log \pi_{\theta}(a_i | s_i)\|. \quad (2)$$

The first factor is sample badness or severity; the second is learner-relative policy geometry. Sample count, directional coherence, and the network Jacobian determine how individual terms aggregate. Equation 2 motivates matching badness while changing only distance or rarity.

We use $r_{\theta}(s, a)$ as a family-indexed remoteness coordinate. For a Gaussian policy, r_{θ} is a registered standardized or Mahalanobis distance (or an equivalent calibrated negative log-density). For a categorical policy, it is action surprisal $-\log \pi_{\theta}(a | s)$; for sequence policies, the registered analogue is normalized token- or completion-level negative log-likelihood. The notation is shared, but the score-growth law is not assumed to be identical across policy families. Crucially, far field is dynamic: a fixed historical action can become remote as the current learner changes.

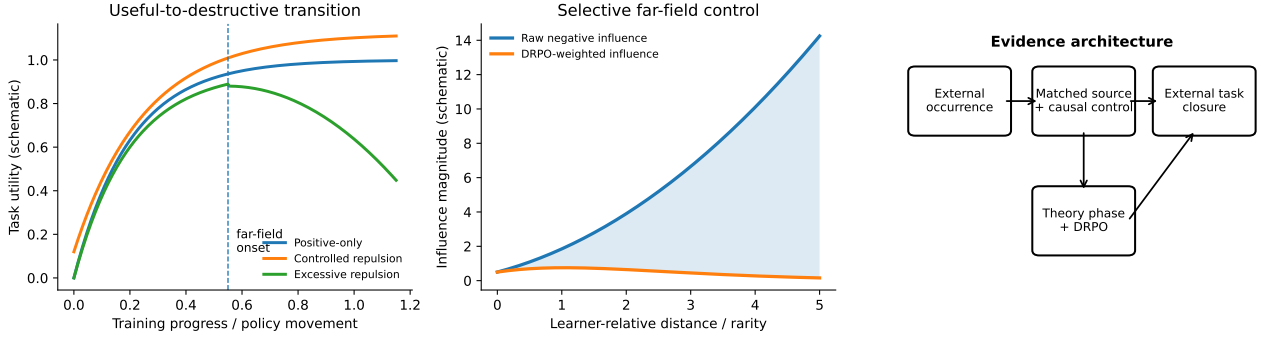


Figure 1. Paper overview. Negative feedback can move a policy beyond the Positive-only target, but historical reuse can turn local feedback into destructive far-field influence. DRPO attenuates that tail. The evidence chain first establishes external occurrence, then performs matched source and causal identification, and finally returns to external task closure. All curves are conceptual; empirical figures are inserted only after formal terminal-audited completion.

4. Repulsive Dynamics

For a Gaussian policy with mean μ and covariance Σ ,

$$\nabla_{\mu} \log \pi(a | s) = \Sigma^{-1}(a - \mu), \quad (3)$$

so the mean-score magnitude grows with standardized distance. When covariance is learned, the variance direction depends on standardized location: the dangerous far-field chain is mean repulsion together with support contraction, not universal expansion of both mean and variance. For categorical logits z ,

$$\nabla_z \log \pi(a | s) = e_a - \pi(\cdot | s). \quad (4)$$

This direct-logit score is bounded, but repeated negative updates continue to decrease the selected action’s log-odds. The shared phenomenon is persistent movement toward a policy-family boundary, not an identical Euclidean gradient law.

Consider a regular minimal exponential-family policy

$$\pi_{\eta}(a) = h(a) \exp\{\eta^T T(a) - \psi(\eta)\}. \quad (5)$$

Let $p, q \geq 0$ denote aggregate positive and negative update mass, and let $\mathbf{m}_+$ and $\mathbf{m}_-$ denote their normalized sufficient-statistic moments. The population signed field is

$$\mathbf{F}(\eta) = p\mathbf{m}_+ - q\mathbf{m}_- - (p - q)\nabla\psi(\eta). \quad (6)$$

These quantities define the aggregate positive–negative competition analyzed by the equilibrium theorem.

Theorem 4.1 (Stable extrapolation and loss of finite equilibrium). *Assume $p > q$ in (6). Define the signed target*

$$\mathbf{m}^* = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p - q}. \quad (7)$$

If $\mathbf{m}^$ lies in the interior of the mean-parameter space, there is a unique finite equilibrium η^* satisfying $\nabla\psi(\eta^*) = \mathbf{m}^*$.*

Moreover,

$$\mathbf{m}^* - \mathbf{m}_+ = \frac{q}{p - q}(\mathbf{m}_+ - \mathbf{m}_-), \quad (8)$$

so negative repulsion moves the equilibrium beyond the Positive-only target in the direction away from the negative moment. As $\mathbf{m}^*$ approaches the feasible boundary, the corresponding natural parameter becomes degenerate or unbounded; if the signed target lies outside the feasible mean-parameter space, no finite equilibrium exists. At an interior equilibrium, $J_F(\eta^*) = -(p - q)\nabla^2\psi(\eta^*)$ is negative definite.

The theorem gives one phase sequence: Positive-only, stable extrapolation, boundary approach, and loss of finite equilibrium. The experiments align this phase classification with separate measurements of task performance, policy-family boundaries, and numerical state.

Theorem 4.1 yields direct experimental predictions. Removing negative updates should produce a stable Positive-only platform. Adding moderate negative influence should move the policy beyond that platform while retaining a finite terminal state. Increasing negative strength or outward moment should approach a covariance, support, or probability boundary; beyond the feasible region, the policy should exhibit persistent drift or a non-vanishing field residual. Finally, selective attenuation of the far-field component should restore a finite regime while retaining part of the extrapolation benefit. We test these predictions in C-U1 and D-U1, reporting task-performance collapse, support or variance boundaries, and NaN/Inf failures as separate outcomes.

For Gaussian policies, an interior signed target corresponds to finite mean and covariance parameters; far-field negative updates can increase standardized displacement and contract support, causing score amplification and boundary approach. For categorical policies, an interior target corresponds to finite logits, whereas a boundary target drives

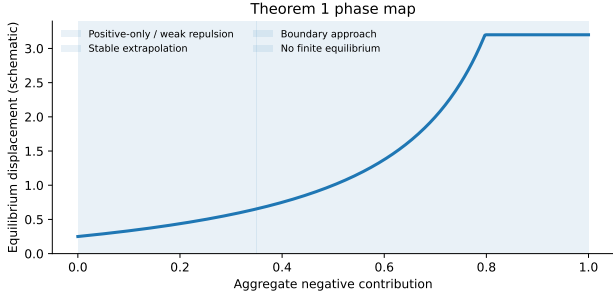


Figure 2. Schematic phase map induced by Theorem 4.1. The empirical tests separately identify the Positive-only target, stable extrapolation, boundary approach, and persistent drift or loss of a finite equilibrium.

selected probabilities toward zero or one. Direct-logit score norms remain bounded, but suppression persists. Thus the transferable principle is aggregate repulsion toward a feasibility boundary, while the local amplification law remains policy-family specific.

5. Distributionally Robust Policy Optimization

DRPO applies a policy-relative envelope to the negative component of the signed actor field. Let $r_\theta(s, a) \geq 0$ denote the registered remoteness measure for the policy family and experiment. We define

$$w_\lambda^-(s, a) = \exp[-\lambda r_\theta(s, a)], \quad \lambda \geq 0, \quad (9)$$

and optimize

$$\mathbf{F}_{\text{DRPO}}(\theta) = \mathbb{E}_\nu [A^+ \nabla \log \pi_\theta - A^- w_\lambda^- \nabla \log \pi_\theta]. \quad (10)$$

Positive updates are unchanged, nearby negative actions retain substantial influence, and increasingly remote negative actions are attenuated. Sample quality and learner-relative remoteness remain distinct axes: the matched experiments vary remoteness without changing badness, while Appendix H states the separate Optimistic-DRO quality-selection result. The main DRPO update does not equate the two controls.

In the exponential-family analysis, the uncontrolled negative moment is

$$q\mathbf{m}_- = \mathbb{E}[A^- T(a)]. \quad (11)$$

DRPO replaces it by

$$q_\lambda \mathbf{m}_{-, \lambda} = \mathbb{E}[A^- e^{-\lambda r_\theta(s, a)} T(a)], \quad (12)$$

which induces the controlled target

$$\mathbf{m}_\lambda^* = \frac{p\mathbf{m}_+ - q_\lambda \mathbf{m}_{-, \lambda}}{p - q_\lambda}. \quad (13)$$

The method therefore modifies the same term that first creates stable extrapolation and later drives the signed target

toward the boundary. Small r_θ preserves substantial negative influence; large r_θ is selectively attenuated.

Proposition 5.1 (Vanishing weighted far-field gradient). *Suppose the policy score satisfies $\|\nabla_\theta \log \pi_\theta(a | s)\| \leq C(1 + r_\theta(s, a))^k$ for finite C, k . For any $\lambda > 0$,*

$$e^{-\lambda r_\theta(s, a)} \|\nabla_\theta \log \pi_\theta(a | s)\| \rightarrow 0 \quad \text{as } r_\theta(s, a) \rightarrow \infty. \quad (14)$$

The exponential form is therefore a gradient-tail envelope derived from the far-field score-growth bound. Directional utility is measured separately in the controlled experiments.

We compare DRPO against an uncontrolled signed baseline, Positive-only learning, non-selective global scaling, registered linear or reciprocal tapers, and a hard threshold on the same remoteness coordinate. A quality-based hard filter is reported separately because it controls a different axis. Where the protocol calls for budget matching, methods receive the same pre-optimizer raw negative-gradient norm and are evaluated with paired seeds. No method ranking is assumed in advance; best and terminal checkpoints, local-negative retention, far-field retention, task performance, boundary events, and numerical failures are all reported.

6. Experiments

We use two primary controlled environments and two external families. C-U1 is a nonlinear state-conditioned Gaussian environment with six-dimensional numerical contexts, two-dimensional actions, a known state-dependent optimum, and independently sampled train and test contexts from the same distribution. It supports matched quality-distance probes, near/far intervention, and mean-support terminal audits. D-U1 is a shared-network categorical environment with unordered semantic actions and a known reward structure; it supports matched quality-rarity probes, common/rare intervention, and probability-boundary audits. We report C-U1 using the precise term held-out-context or unseen-state generalization. Hopper/D4RL adds a learned critic and long-horizon continuous control, while Countdown adds shared Transformer parameters and sequence-level verification. External tasks establish relevance; controlled environments establish source and causality.

Hopper/D4RL. We train the registered learned critic, freeze its selected checkpoint, and audit negative actor gradients by current-policy distance. The formal report will include far/near negative-gradient ratios, positive/negative imbalance, policy drift, rollout return, and terminal classification. **TBD:** insert only terminal-audited formal results from the registered E7 package.

Countdown. We bin negative tokens or completions by current-policy surprisal and track score magnitude, shared-

Table 1. Evidence roles are separated by design. Controlled environments identify source and causality; external environments establish relevance and task effect.

Environment	Primary responsibility	Complementary boundary
C-U1	Continuous matched source isolation, near/far causality, and phase transition	External relevance is established in Hopper/D4RL; categorical boundaries are tested in D-U1
D-U1	Categorical matched rarity isolation, common/rare causality, and probability boundary	Gaussian score-amplitude dynamics are tested in C-U1
Hopper/D4RL	External continuous-control occurrence and task effect with a learned critic	Exact badness-distance isolation is supplied by C-U1
Countdown	External shared-parameter sequence occurrence and task effect	Controlled common/rare causality is supplied by D-U1

Table 2. D-U1 E5 long-run summary.

Method	Reward	Task collapse	Boundary
baseline	0.001	20/20	20/20
near-zero	0.006	20/20	20/20
far-zero	0.267	0/20	0/20
far-cap	0.297	0/20	0/20
global-scale	0.957	0/20	20/20
positive-only	0.275	0/20	0/20

Table 3. C-U1 intervention summary (20 seeds).

Method	Reward	Task event	Boundary event
baseline	0.603	0.00	1.00
near-zero	0.602	0.00	1.00
far-zero	0.653	0.00	0.00
far-cap	0.653	0.00	0.00
global-scale	0.643	0.00	0.00

parameter gradient influence, target-probability change, verifier success, validity, and pass@ k . **TBD**: insert only the registered formal result; focused pilots remain provenance rather than final evidence.

The source-isolation protocol holds sample badness fixed. In C-U1, negative actions share the same state, reward, advantage magnitude, quality coordinate, count, coefficient, and policy parameters while varying only learner-relative radius. We report output-space score norms, full-parameter per-sample gradients, aggregate gradients, and directional coherence. In D-U1, common and rare actions are matched in semantic role, reward, negative advantage, count, and base coefficient while initial probability or current surprisal changes. A far/near or rare/common gap under these controls identifies policy remoteness as an independent amplifier; it does not imply that distance is the only factor in real data.

We next intervene on the location of negative influence. The continuous comparison includes the uncontrolled signed baseline, Near-zero, Far-zero, Far-cap, a non-selective global control matched to the retained raw negative-gradient budget, and a registered budget-transfer control. The categorical analogue suppresses common or rare negatives under matched counts and budgets. We record the onset order of remote influence, parameter drift, support or probability boundaries, and task performance. A selective far/rare rescue together with an ineffective near/common removal supports the remote component as a causal transmission pathway; task-performance collapse, boundary events, and NaN/Inf remain distinct labels.

We sweep the effective negative strength from zero through controlled repulsion and into the unstable regime. The primary test is not a single endpoint score but the phase sequence predicted by Theorem 4.1: Positive-only, stable extrapolation, boundary approach, and persistent drift or loss of finite equilibrium. We measure a family-specific empirical proxy for the aggregate negative term in (12), together with its distance or rarity bins, direction, and relationship to equilibrium displacement. DRPO is compared with Positive-only, global scaling, and registered distance controls under paired seeds and explicit raw-gradient budgets. Results are reported at both selected and terminal checkpoints. **TBD**: populate the final table only from the current terminal-audited repository artifacts and preserve each result’s formal status.

The final evaluation returns to external tasks. On D4RL locomotion, methods share dataset, critic protocol, initialization, seeds, evaluation horizon, and selection rule; we report normalized return, uncertainty, best and terminal checkpoints, and distance-binned negative influence. On Countdown, methods share the SFT initialization, rollout or replay bank, verifier, seeds, and checkpoint-selection protocol; we report greedy success, pass@ k , validity, terminal degradation, and rare-negative diagnostics. **TBD**: the main external tables remain unfilled until the corresponding registered formal experiments are terminal-audited and delivered.

7. Implications and Conclusion

Negative feedback is a policy-improvement resource with a dynamical failure mode. It provides boundary information and can shift a policy beyond the Positive-only target. The

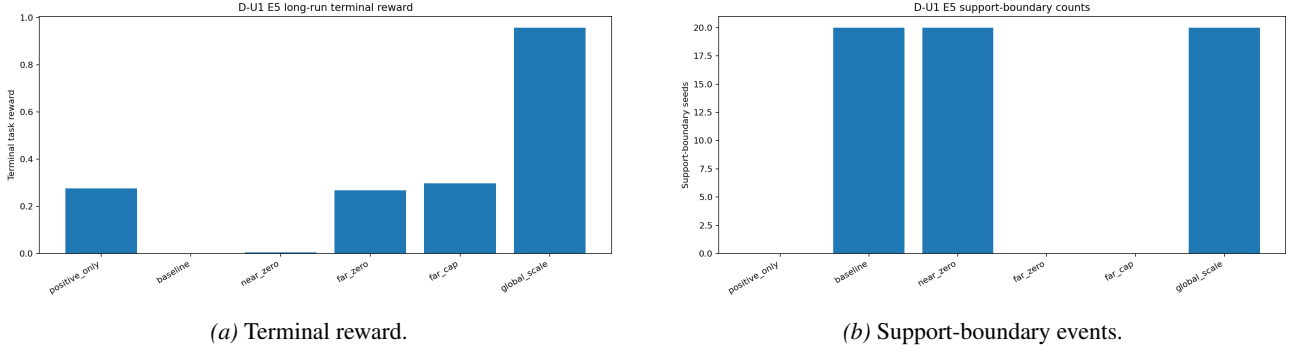


Figure 3. D-U1 long-run causal results. Baseline and Near-zero preserve the rare/far negative pathway and reach the support boundary in 20/20 seeds; Far-zero and Far-cap remain bounded in 20/20 seeds. Global scaling can preserve task reward while still reaching the support boundary, illustrating why task, boundary, and numerical events must be reported separately.

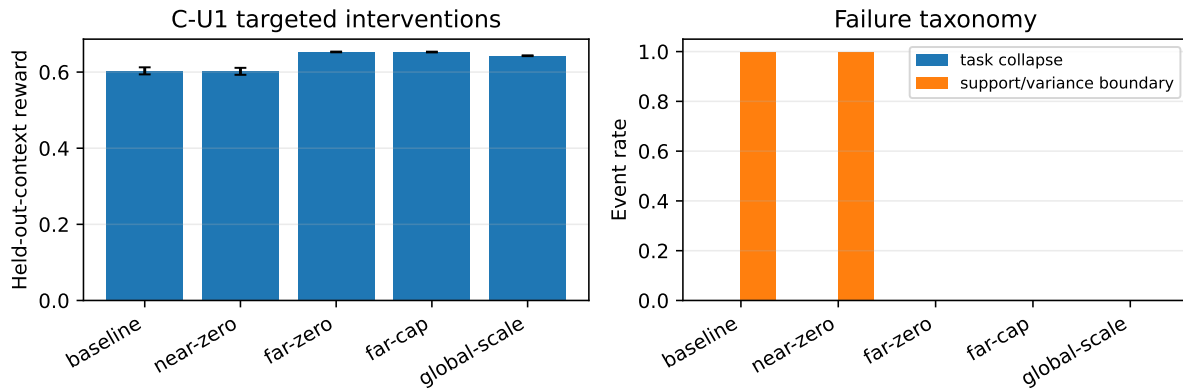


Figure 4. C-U1 causal intervention with learnable variance. The left panel reports registered held-out-context reward with confidence intervals; the right panel separates task-performance collapse from support or variance-boundary events. Near-zero tracks the uncontrolled baseline, while Far-zero and Far-cap remove the task-collapse pathway without introducing NaN/Inf failures.

failure arises when historical actions remain active after becoming remote from the current learner, so their optimization influence no longer tracks their local relevance. The transferable design principle is therefore to control learner-relative far-field negative influence rather than delete negative feedback as a class.

The continuous and categorical analyses share an aggregate structure but not an identical local law. Gaussian policies can amplify mean scores with standardized distance and couple repulsion to support contraction. Categorical direct-logit scores remain bounded, yet repeated negative updates can persistently suppress probability. In both cases, aggregate positive-negative competition determines whether the policy remains at an interior equilibrium or moves toward a feasibility boundary.

Negative feedback can create stable policy improvement beyond Positive-only learning. By separating badness from policy remoteness, we identify how historical negative actions become far-field, dominate the aggregate update, and move the policy toward a feasibility boundary where a fi-

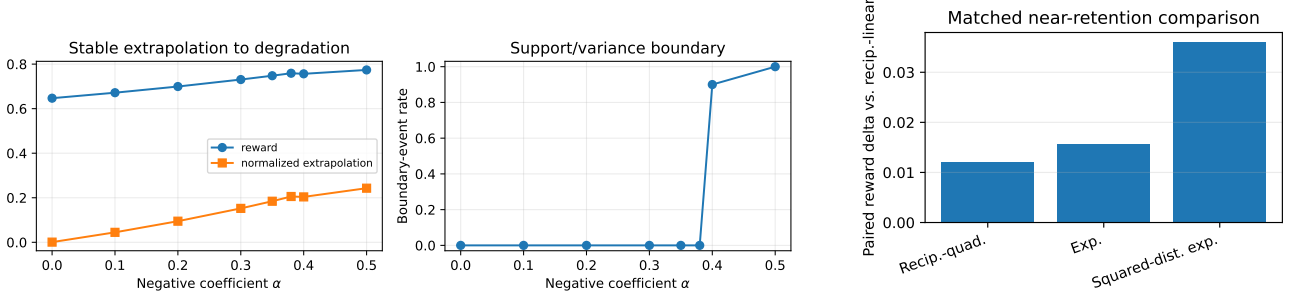
nite equilibrium can disappear. DRPO controls this same negative term, preserving useful local repulsion while suppressing the destructive far-field tail.

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Table 4. Controlled phase and method results. Entries remain TBD until the corresponding registered artifacts and terminal audits are frozen for the manuscript. Task collapse, boundary events, and numerical failure are reported separately.

Method	Held-out-context reward	Near retention	Far retention	Task collapse	Boundary event	NaN/Inf
Positive-only	TBD	—	0	TBD	TBD	TBD
Uncontrolled signed	TBD	TBD	TBD	TBD	TBD	TBD
Global matched	TBD	TBD	TBD	TBD	TBD	TBD
Hard distance control	TBD	TBD	TBD	TBD	TBD	TBD
DRPO exponential	TBD	TBD	TBD	TBD	TBD	TBD



(a) Strength sweep and boundary events.

(b) Matched near-retention taper comparison.

Figure 5. C-U1 phase and control evidence. The strength sweep maps the transition from the Positive-only reference through stable extrapolation toward boundary events. Under the frozen near-retention protocol, all three faster-decaying tapers improve finite-horizon held-out-context reward on 20/20 paired seeds relative to reciprocal-linear; this result remains finite-step validated rather than a universal steady-state ranking.

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Table 5. External task closure. Best and terminal results are both required; no values are inserted before formal terminal-audited completion.

Family	Dataset/model	Uncontrolled	Positive-only	Global	DRPO	Terminal failure rate
D4RL	Hopper / locomotion suite	TBD	TBD	TBD	TBD	TBD
Countdown	Main model scale	TBD	TBD	TBD	TBD	TBD

A. Proofs for Repulsive Dynamics

For a regular minimal exponential family, $\nabla\psi$ maps the natural-parameter space diffeomorphically onto the interior of the mean-parameter space. Setting (6) to zero with $p > q$ gives (7). If $\mathbf{m}^*$ is interior, uniqueness follows from strict convexity of ψ . The displacement identity follows by subtracting $\mathbf{m}_+$ from (7). If $\mathbf{m}^*$ approaches the boundary, no finite natural parameter realizes it; if it lies outside the closure, no policy in the family realizes the target. Differentiating (6) gives $J_F = -(p - q)\nabla^2\psi$, which is negative definite at an interior point because the Hessian is the positive-definite covariance of the sufficient statistic. The discrete update is locally stable for step sizes whose transition matrix has spectral radius below one.

A.1. Detailed derivation for Theorem 4.1

Proof. Let $F(\eta) = p(\mathbf{m}_+ - \nabla\psi(\eta)) - q(\mathbf{m}_- - \nabla\psi(\eta))$. Rearranging gives

$$F(\eta) = p\mathbf{m}_+ - q\mathbf{m}_- - (p - q)\nabla\psi(\eta). \quad (15)$$

For $p > q$, $F(\eta) = 0$ is equivalent to

$$\nabla\psi(\eta) = \mathbf{m}^* = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p - q}. \quad (16)$$

A regular minimal exponential family has strictly convex log-partition ψ , so $\nabla\psi$ is one-to-one and maps the natural-parameter space onto the interior of the realizable mean-parameter set. Hence an interior $\mathbf{m}^*$ has a unique finite preimage η^* . Subtracting $\mathbf{m}_+$ yields

$$\mathbf{m}^* - \mathbf{m}_+ = \frac{q}{p - q}(\mathbf{m}_+ - \mathbf{m}_-), \quad (17)$$

which is the stable extrapolation displacement. Differentiating the field gives

$$J_F(\eta) = -(p - q)\nabla^2\psi(\eta). \quad (18)$$

At η^* , $\nabla^2\psi$ is the covariance of the sufficient statistic and is positive definite; therefore every eigenvalue of J_F is negative and the continuous-time equilibrium is locally asymptotically stable. For discrete step size h , local stability requires $\rho(I + hJ_F) < 1$. If $\mathbf{m}^*$ reaches the feasible boundary, the realizing natural parameter diverges or becomes degenerate; outside the feasible set, no finite equilibrium exists. $\square$

A.2. Proof of Proposition 5.1

Proof. Assume the unweighted score satisfies $\|\nabla_\theta \log \pi_\theta(a \mid s)\| \leq C(1 + r_\theta)^k$ for finite k , and let $w_\lambda(r) = \exp(-\lambda r^2)$. Then

$$\|w_\lambda(r)A^\top \nabla_\theta \log \pi_\theta\| \leq |A^\top|C(1 + r)^k e^{-\lambda r^2}. \quad (19)$$

For every finite k and $\lambda > 0$, the Gaussian tail dominates the polynomial factor, so the right-hand side converges to zero as $r \rightarrow \infty$. This proves vanishing weighted far-field influence without assuming that the sample's utility decays exponentially. $\square$

B. Gaussian Mean–Variance Derivations

For a scalar Gaussian with $\xi = \log \sigma$ and $z = (a - \mu)/\sigma$,

$$\partial_\mu \log \pi = \frac{a - \mu}{\sigma^2}, \quad \partial_\xi \log \pi = z^2 - 1. \quad (20)$$

A negative advantage reverses both score directions. It always pushes the mean away from the action, but its effect on scale depends on $|z|$: a near negative action with $|z| < 1$ increases σ , whereas a far negative action with $|z| > 1$ decreases σ . Positive updates have the opposite signs. A full Gaussian terminal state must satisfy both mean and variance equations; a stationary mean with continuing covariance movement is not a complete policy equilibrium.

B.1. Population mean–variance fixed point

With $\xi = \log \sigma$, positive mass p , effective negative mass q , conditional moments (m_+, v_+) and (m_-, v_-) ,

$$\dot{\mu} = \frac{p(m_+ - \mu) - q(m_- - \mu)}{\sigma^2}, \quad (21)$$

$$\dot{\xi} = \frac{p[v_+ + (\mu - m_+)^2] - q[v_- + (\mu - m_-)^2]}{\sigma^2} - (p - q). \quad (22)$$

For $p > q$, the mean candidate is $\mu^* = (pm_+ - qm_-)/(p - q)$. Substitution into the scale equation gives

$$(\sigma^2)^* = \frac{pM_+(\mu^*) - qM_-(\mu^*)}{p - q}, \quad (23)$$

where $M_\pm(\mu) = v_\pm + (\mu - m_\pm)^2$. A finite joint equilibrium therefore requires both $p > q$ and a positive numerator. This second condition is stricter in the far field because M_- contains squared displacement.

C. Categorical Boundary Dynamics

For softmax probabilities p_j , the score for selected action y is $e_y - p$, whose Euclidean norm is bounded. Under a negative update with fixed local coefficient, the selected logit decreases relative to competing logits, so $z_y - z_j$ falls approximately linearly while p_y can decay exponentially. The failure mode is therefore persistent support suppression toward the simplex boundary. In a shared network, the parameter-space gradient additionally contains the network Jacobian and cross-action interference, which are measured rather than inferred from the direct-logit bound alone.

C.1. Repeated negative updates and log-odds

For softmax logits z , $\nabla_z \log p_y = e_y - p$. Under a fixed negative coefficient $c > 0$ and small step size h , the selected logit changes by $-hc(1 - p_y)$ while competitor j changes by $+hcp_j$. Hence

$$(z_y - z_j)_{t+1} - (z_y - z_j)_t = -hc(1 - p_y + p_j), \quad (24)$$

which remains negative as $p_y \rightarrow 0$. The Euclidean score is bounded, but the non-vanishing log-odds decrement yields persistent support suppression and can drive p_y exponentially toward the simplex boundary.

D. Controlled Environments

C-U1 samples numerical contexts $s \sim \mathcal{N}(0, I_6)$ independently for training and testing. A fixed generator maps each state to a positive action, hidden optimal action, negative quality coordinate, and reward surface in $\mathbb{R}^2$. Source-isolation probes replicate reward and advantage across radii while retaining the same state and quality coordinate. Near/far membership for causal interventions is computed relative to the current policy according to the registered metric. D-U1 uses a shared state-conditioned categorical network over unordered semantic actions; quality/semantics and rarity are factorized so that common/rare probes can be matched in reward, advantage, count, and coefficient. Complete constants, seeds, and invariants are taken from the registered experiment configurations.

E. Experimental Protocols and Terminal Audits

Formal protocols specify maximum horizons, evaluation cadence, paired development and held-out seeds, and a terminal audit before execution. A fixed horizon is not itself called convergence. Depending on the registered experiment, terminal classification uses state slopes, update residuals, boundary checks, and a continuation horizon. Budget-matched comparisons use the pre-optimizer raw negative-gradient ℓ_2 norm unless another coordinate is explicitly frozen. Adam parameter-update norms are recorded separately. Best-validation and terminal checkpoints are both reported, and task-performance collapse, support or variance boundaries, and NaN/Inf failures remain distinct fields.

F. Additional Results and Failure Taxonomy

This appendix will contain per-seed results, full training trajectories, confidence intervals, sensitivity analyses, and terminal classifications for every main comparison. Failed or inconclusive formal runs are indexed rather than silently removed. **TBD:** populate these materials from the durable

formal packages after repository closure; smoke tests, static checks, and focused pilots are not promoted to formal evidence.

G. Implementation and Reproducibility

The repository records experiment IDs, configurations, seeds, code entry points, result manifests, raw curves, logs, terminal audits, checksums, and source commits. Formal runs progress through registered, running, raw-complete, terminal-audited, packaged, delivered, and applied states. The paper source is generated from the stable-ID manuscript graph, compiled in the imported ICML/Overleaf project, and packaged with a reproducible build command. Exact repository paths and commit hashes will be inserted at submission freeze.

H. Optimistic Distributional Selection

Let P denote the empirical data distribution and let $R(s, a)$ be the registered quality signal. The Optimistic-DRO selection problem is

$$\begin{aligned} \max_Q \quad & \mathbb{E}_Q[R(s, a)] \\ \text{s.t.} \quad & Q \ll P, \quad 0 \leq \frac{dQ}{dP} \leq \frac{1}{\kappa}, \\ & \int dQ = 1. \end{aligned} \quad (25)$$

Its optimum allocates the maximum admissible density to the highest-quality region of P , with boundary mass only when needed to satisfy normalization. For a continuous reward distribution this is the top- κ hard-selection solution. This result establishes a principled quality-selection operator inside the current DRPO manuscript. It does not identify policy remoteness with low quality and does not derive the exponential envelope in Eq. (9). Quality selection controls which data are emphasized; remoteness tapering controls how a negative sample’s influence changes as the learner moves. The source-isolation experiments keep quality fixed precisely so that the second effect can be identified independently.

I. Additional Theoretical and Reporting Clarifications

The manuscript uses the following technical conventions throughout. Far-field negative Gaussian updates combine mean repulsion with location-dependent scale dynamics and commonly support contraction; they do not universally expand both mean and variance. The expected Fisher matrix describes policy geometry but does not by itself determine stability of a fixed off-policy signed field; local stability is evaluated with the Jacobian of the actual aggregate field.

C-U1 test states are independently sampled from the same state distribution and are described as held-out-context or unseen-state generalization, not OOD generalization. Task-performance collapse, support or variance boundaries, and NaN/Inf numerical failure are reported separately. Finally, finite-horizon evidence is not relabeled as convergence or long-run validation without the registered terminal audit.